

Conformalized Zero-Shot Forecasting

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Abstract

Time Series Foundation Models (TSFMs) have transformed the time series prediction landscape by enabling zero-shot forecasting (ZSF) with accurate point predictions and even native uncertainty quantification via quantile outputs. By rigorously evaluating against post-hoc conformal prediction (CP) calibration, we show that these quantile outputs can be poorly calibrated. We provide a standardized application of and comprehensive comparison between CP variants in the ZSF setting, and demonstrate that they can significantly improve coverage over native TSFM quantiles.

1 Introduction

Time series forecasting is a critical task across various domains, including finance, healthcare, weather prediction, and supply chain management, where accurate future predictions drive essential decision-making. Traditionally, forecasting has relied on statistical methods such as ARIMA or Exponential Smoothing, as well as machine learning approaches like Gradient Boosting or recurrent neural networks (RNNs) trained specifically on individual datasets. While effective for well-defined problems, these traditional methods often struggle to generalize across diverse domains and require extensive dataset-specific tuning.

Recently, the paradigm has shifted with the introduction of Time Series Foundation Models (TSFMs). Inspired by the success of large language models, TSFMs are pre-trained on massive, diverse datasets, enabling them to capture universal temporal patterns. Models such as TimesFM, Chronos, and FlowState have demonstrated highly competitive point prediction performance, often achieving state-of-the-art results in zero-shot settings without the need for task-specific fine-tuning.

However, in many real-world applications, point predictions are insufficient. Uncertainty quantification (UQ) is often useful or even strictly required to assess the reliability of a forecast and mitigate risks. Even though most modern TSFMs now offer native UQ capabilities, such as outputting predictive distributions or utilizing quantile regression, these estimates generally come with no formal statistical guarantees. They can be overly confident or poorly calibrated, especially when applied to unseen data distributions in a zero-shot manner.

To bridge this gap, Conformal Prediction (CP) offers a powerful, distribution-free framework for rigorous uncertainty quantification. By constructing prediction intervals targeting a user-specified probability of containing the true future values, CP helps restore reliability to TSFM forecasts. In this work, we explore the integration of time-series-specific conformal prediction methods with state-of-the-art TSFMs, aiming to combine the strong zero-shot predictive power of foundation models with the robust, mathematically motivated coverage properties of conformal inference.

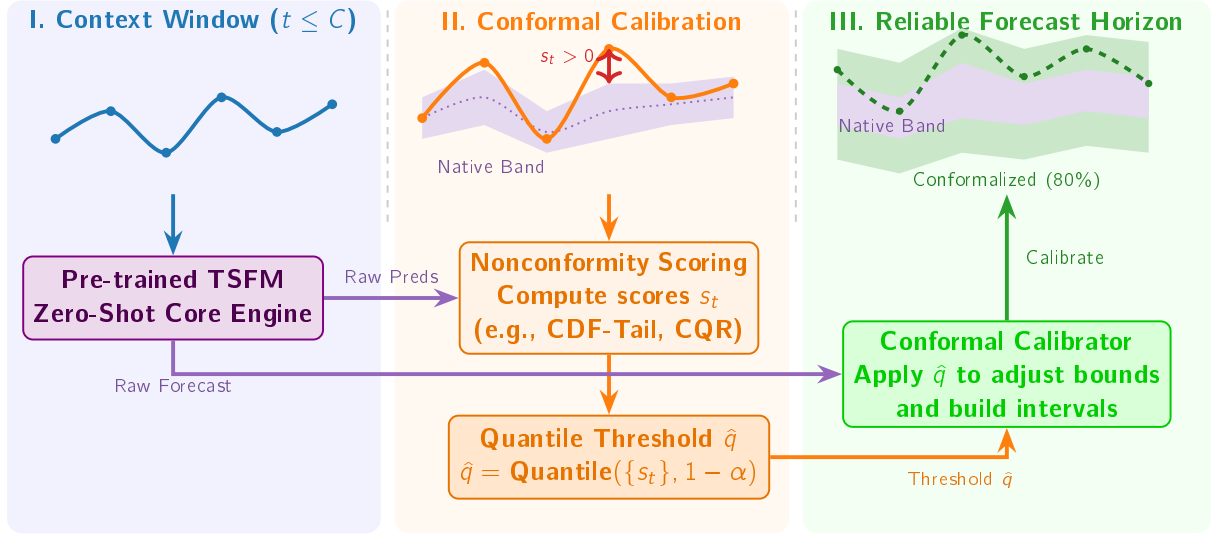


Figure 1: Overview of the Conformalized Zero-Shot Forecasting Pipeline. (I) The pre-trained Time Series Foundation Model (TSFM) takes the Context Window as input. (II) Over a disjoint Calibration Set, we compute nonconformity scores (s_t) by comparing the actual values against the TSFM’s native predictions. The empirical $(1 - \alpha)$ quantile of these scores yields a threshold ($\hat{q}$). (III) In the future Forecast Horizon, the Conformal Calibrator scales the TSFM’s native quantiles using $\hat{q}$, producing calibrated prediction intervals with mathematically guaranteed $(1 - \alpha)$ marginal coverage, effectively correcting the base model’s uncalibrated under-coverage.

2 Background and Related Work

2.1 Time Series Foundation Models

The paradigm of foundation models, which has revolutionized natural language processing and computer vision, is increasingly being adapted for time series analysis. Time Series Foundation Models (TSFMs) are typically pre-trained on massive, diverse datasets, allowing them to capture universal temporal patterns and enabling zero-shot or few-shot inference on unseen datasets. These models can be broadly categorized based on their architectural choices and representation strategies.

Patch-based and Generic Transformers Early efforts to scale time series forecasting demonstrated that segmenting time series into sub-sequence patches significantly improves the capability of Transformers to capture local semantics and long-term dependencies. A notable milestone in this direction is PatchTST [WGR⁺26], which leverages channel-independent processing and patching to establish a strong baseline for subsequent foundation models.

Decoder-Only and Language-Model-Based Models Drawing inspiration from large language models, several TSFMs employ decoder-only architectures, often treating time series values as sequences of tokens or patches. TimesFM [DKSZ24] (and its later iterations like TimesFM-2.5) is a decoder-only foundation model explicitly designed for time series forecasting that processes patches of data to maintain competitive inference speed

and accuracy. Chronos-2 [ASK⁺25] is a prominent example that learns the “language” of time series by tokenizing continuous values and training an autoregressive language model, demonstrating strong zero-shot capabilities for universal forecasting. Toto-2 [KLW⁺26] similarly scales autoregressive time series forecasting to massive datasets, highlighting the benefits of scaling laws in the time series domain. Timer-S1 [LSW⁺26] pushes the boundary of sheer scale, introducing a Mixture-of-Experts (MoE) billion-scale time series foundation model that employs serial scaling techniques to handle massive context lengths and complex temporal relationships.

Beyond Transformers While transformer-based and autoregressive token models are powerful, alternative architectures aim to improve forecasting by utilizing continuous representations, recurrent state-tracking, or state space models. Unlike most TSFMs that rely on transformers, TiRex [APK⁺25] leverages xLSTM—an enhanced Long Short-Term Memory network—to retain strict recurrent state-tracking, which is argued to be highly beneficial for robust in-context learning and long-horizon zero-shot forecasting. Conversely, FlowState [GOWP25] departs from discrete patch or token processing by utilizing a state space model (SSM) based encoder coupled with a functional basis decoder. This architectural choice enables continuous-time modeling and dynamic time-scale adjustments, providing a sampling-rate invariant formulation that generalizes effectively across frequencies without needing to memorize patterns at every scale. Finally, returning to the transformer paradigm,

Multi-step Forecasting It is important to note that current Time Series Foundation Models mostly perform direct multi-step forecasting. Unlike traditional autoregressive models that predict one step at a time, TSFMs take a historical context window as input and directly output a sequence of future predictions over a specified forecasting horizon.

2.2 Conformal Prediction

Conformal prediction (CP) [VGS05] is a versatile, distribution-free framework for uncertainty quantification. For a comprehensive overview of the CP framework from a data-centric perspective, we refer the reader to [ZCGC25]. Given a user-specified error rate or *miscoverage rate* $\alpha \in (0, 1)$, CP constructs prediction intervals (or sets) that are mathematically guaranteed to contain the true test response with probability at least $1 - \alpha$. This theoretical guarantee, known as *marginal coverage*, holds regardless of the underlying data distribution or the chosen base predictive model, provided that the data points are exchangeable (i.e., the joint probability distribution is invariant to permutations of the indices).

The calibration process in the standard split CP setup proceeds as follows. First, the available data is divided into a training set and a disjoint *calibration set*. A base machine learning model is trained on the training set. Then, a heuristic notion of uncertainty is encapsulated into a *non-conformity score*, which measures how unusual a new observation is relative to the model’s predictions. These scores are computed for all instances in the calibration set. By calculating the empirical $\frac{[(n+1)(1-\alpha)]}{n}$ quantile of these calibration non-conformity scores (where n is the size of the calibration set), CP determines a critical threshold. Finally, for any new test point, the prediction interval is constructed by including

all possible target values whose non-conformity score falls below this calibrated threshold. This elegantly simple procedure calibrates the output of any base model into valid, theoretically sound prediction intervals without making strong distributional assumptions.

2.2.1 Methods for Time Series Data

Applying standard split CP to time series data presents significant theoretical and practical challenges. The primary obstacle is the violation of the exchangeability assumption. Time series data exhibit temporal dependencies, autocorrelations, and non-stationarity (such as concept drift, seasonality, and varying volatility). Consequently, the non-conformity scores evaluated on past data may not reliably represent the distribution of scores for future observations. When standard CP is applied naively to time series, it often results in severe under-coverage or unnecessarily wide, uninformative prediction intervals. Thus, specialized adaptations of the CP framework are required to dynamically adjust to changing environments and shifting distributions.

To address the lack of exchangeability, a growing body of literature has developed conformal prediction methods specifically tailored for time series data. These methods can be broadly categorized into the following groups:

Adaptive and Online Conformal Inference One prominent model-agnostic approach is to adapt the target error rate or the interval width iteratively based on past coverage performance. For a gentle, tutorial-style introduction to conformal forecasting methods in sequential and time-series environments, see [SFT⁺25]. If recent intervals have been under-covering the true values, the algorithm widens the intervals for subsequent predictions; conversely, it narrows them if over-coverage is detected. Adaptive Conformal Inference (ACI) [GC21] and its variations AgACI [ZFG⁺22] and DtACI [GC24] follow this principle. More recent works have formulated this adaptation as a dynamical system control problem, introducing Conformal PID Control [ACT23], Neural Conformal Control [LR25], and online conformal prediction with decaying step sizes [ABB24] to achieve robust sequential coverage.

Handling Covariate Shift and Change Points A distinct line of research focuses on adapting conformal prediction when the underlying distribution exhibits explicit shifts or abrupt structural changes. Early work modified the CP procedure by reweighting non-conformity scores to account for covariate shift [TFBCR19]. Later, general theoretical frameworks were developed to understand conformal prediction beyond exchangeability [BCRT23], quantifying the coverage gap induced by such distribution shifts. When facing highly non-stationary environments, specialized methods have also been introduced to maintain valid coverage in the presence of abrupt change points [SY25].

Ensemble and Sequential Calibration Other methods relax the exchangeability assumption by utilizing robust ensemble techniques and sequential calibration. For instance, EnbPI (Ensemble prediction intervals) [XX21] constructs sequential prediction intervals around bootstrap ensemble estimators without requiring data exchangeability, assuming instead that the errors are strongly mixing. Building upon this, Sequential Predictive Conformal Inference (SPCI) [XX23] leverages sequential non-conformity scores and updates

the calibration set dynamically to better capture local temporal dynamics. EnCQR [JBA24] extends these ideas to quantile regression, constructing prediction intervals around ensemble quantile regression models. While powerful, these ensemble approaches require training multiple models on subsets of data, which can be computationally intensive and does not align well with the concept of foundation models that are designed for zero-shot inference.

Conformalized Quantile Regression and Temporal Adjustments Combining conformal prediction with quantile regression allows the construction of intervals that naturally adapt their width to heteroscedasticity. Conformalized Quantile Regression (CQR) [RPC19] applies standard CP to the lower and upper quantiles of a base model to obtain adaptively scaled intervals. Various formulations and comparative analyses of conformalized quantile regression methods have been explored across deep learning and random forest backbones [SC20, KJL20]. Additionally, adjusting non-conformity scores through Temporal Quantile Adjustments [LTS22] helps in capturing varying temporal uncertainties, resulting in tighter and more reliable intervals.

2.2.2 Multi-step Forecasting

While many methods focus on one-step-ahead forecasting, extending CP to multi-step horizons requires capturing joint distributions over future time steps. CF-RNN [SMAVdS21] addresses the exchangeability collapse by dividing the data into training/calibration/test splits across timeseries rather than across the time axis. Copula Conformal Prediction [SY23] utilizes copulas to model the dependencies among future steps for coherent multi-step intervals. Advanced perspectives such as Bellman Conformal Inference [YCL24] connect conformal prediction to dynamic programming and reinforcement learning to calibrate intervals for sequential decision-making. Multi-dimensional forecasting challenges are addressed by Flow-based Conformal Prediction [LXX25], employing classifier-free guidance to construct tighter prediction sets. Recent work has also explored integrating conformal prediction directly into specialized neural architectures, such as modern Hopfield networks [AGKH23], to leverage their continuous state representations for more robust uncertainty quantification.

For multi-step-ahead adaptive forecasting, several online conformal inference frameworks have been proposed to handle temporal dependencies and bias corrections over multiple future look-ahead horizons [HS24, STM24, YCW⁺25, LJK26, LMR26]. Furthermore, robust calibration over multi-dimensional trajectories and heterogeneous sequential streams has been investigated using ellipsoidal sets and adaptive scaling techniques [XJX24, ZLS24].

2.3 Conformal Prediction for Time Series Foundation Models

The intersection of Conformal Prediction (CP) and Time Series Foundation Models (TSFMs) is an emerging and highly promising area of research. By leveraging the strong zero-shot predictive capabilities of TSFMs, researchers can apply post-hoc uncertainty quantification methods like CP more effectively. The most direct precursor to our work is [ABNC25], where authors demonstrate that the zero-shot nature of FMs allows a larger proportion of available historical data to be allocated strictly for calibration rather than model training. This results in more stable and reliable conformalized prediction intervals,

particularly in data-constrained scenarios where traditional statistical or machine learning models struggle due to limited training samples. However, this work only includes standard split CP as the method of choice instead of more advanced methods that are designed to handle the specific challenges of time series data, and its dataset choice is limited.

Even if previously unexplored, the integration of time series-specific CP methods with TSFMs is a natural next step. For example, adaptive conformal inference techniques could be applied to the outputs of TSFMs to dynamically adjust prediction intervals in response to detected distribution shifts or temporal dynamics. Similarly, methods designed to handle covariate shift and change points could be particularly beneficial when applied to the forecasts of TSFMs, which may encounter varying patterns across different datasets or time periods. One exception is ensemble-based methods like EnbPI and EnCQR, because their demand of training multiple models on subsets of data is inherently in contrast with the concept of TSFMs.

These integrations highlight the significant potential of combining the representational power of foundation models with the rigorous, distribution-free statistical guarantees provided by conformal prediction.

3 Methods and Experimental Setup

3.1 TSFMs

To rigorously evaluate the integration of conformal prediction with modern foundation models, we select a shortlist of four state-of-the-art Time Series Foundation Models (TSFMs): **Chronos-2** [ASK⁺25], **TimesFM** [DKSZ24] (specifically TimesFM 2.5), **TiRex** [APK⁺25], and **FlowState** [GOWP25]. These models were chosen based on four primary criteria: (1) high public adoption and community usage, (2) ease of use and API integration, (3) strong performance on prominent large-scale benchmarks such as GIFT-Eval [AWL⁺25], TIME [QPW⁺26], and FEV [SAT⁺25], and (4) architectural diversity.

Architecturally, the shortlisted models represent distinct paradigms:

- **TimesFM** and **Chronos-2** are popular transformer-based designs representing decoder-only and encoder-only designs respectively.
- **TiRex** represents an LSTM-based paradigm, utilizing the advanced xLSTM architecture to achieve enhanced recurrent state-tracking and in-context learning.
- **FlowState** represents a state-space model (SSM) based architecture, employing a continuous-time SSM encoder with a functional basis decoder.

By choosing models spanning transformer, recurrent (LSTM), and state-space (SSM) designs, we can evaluate how post-hoc conformal prediction calibrates forecasts across fundamentally different underlying architectures.

3.2 Datasets

To evaluate the robustness, generalization, and calibration of conformalized TSFMs, we identify four major time-series meta-datasets: **GIFT-Eval** [AWL⁺25], **TIME** [QPW⁺26], **BOOM** [CKD⁺25], and **FEV** [SAT⁺25].

- **GIFT-Eval** [AWL⁺25] is currently the most widespread benchmark for general time series forecasting. However, some of the modern TSFMs include subsets of its constituent datasets in their pre-training corpora, meaning evaluations on it are not truly zero-shot.
- **TIME** [QPW⁺26] directly addresses this pre-training leakage by introducing novel, unseen-before temporal datasets specifically selected to guarantee genuine zero-shot evaluation.
- **BOOM** [CKD⁺25] is a specialized benchmark that focuses on specific industrial sub-domains, most notably IT observability and systems monitoring (e.g., CPU, memory, and network telemetry).

Ultimately, we select the **FEV** (Forecasting Evaluation and Verification) benchmark [SAT⁺25] for our main evaluations. We chose FEV for several reasons: first, it is highly comprehensive, incorporating diverse datasets from both GIFT-Eval and BOOM, among other sources; second, it provides a well-defined small subset of representative tasks (**FEV-bench mini**) which accurately represents the complexity of the full benchmark while remaining computationally tractable; third, we found the FEV framework and task layout to be significantly better documented and more straightforward to use and integrate into our evaluation pipeline compared to other meta-datasets. **FEV-bench mini** is the primary benchmark used in our experiments. Importantly, all multivariate datasets and multi-channel tasks are converted to univariate series, with each channel calibrated and forecasted independently to ensure uniform compatibility across all evaluated pre-trained foundation models.

3.3 Conformal Prediction Methods

We evaluate a representative set of conformal prediction (CP) methods that are well-suited for sequential time-series forecasting. These methods are post-hoc calibrators that take point forecasts and predictive distributions produced by TSFMs and generate mathematically sound prediction intervals. The evaluated methods are categorized as follows:

- **Naive Split CP** [PPVG02, VGS05]: A standard, static baseline that assumes exchangeability of nonconformity scores and does not perform online updates.
- **Weighted CP** [TFBCR19]: A baseline that uses decaying geometric weights over past nonconformity scores to prioritize recent observations, offering valid coverage guarantees under certain non-exchangeable distribution shifts.
- **Trailing Window** [BCRT23]: A simple rolling-quantile method that computes empirical quantiles over a sliding window of the most recent scores without explicit error-rate tracking.
- **Adaptive Conformal Inference (ACI)** [GC21]: An online method that dynamically updates the target miscoverage rate α_t at each step based on whether the previous forecast fell within the prediction interval.

- **AgACI** [ZFG⁺22] and **DtACI** [GC24]: Robust expert-ensemble extensions of ACI that self-tune the step-size parameter γ using online learning algorithms (such as AdaHedge and exponential weights) to handle arbitrary distribution shifts.
- **Conformal PID Control (PID)** [ACT23]: Formulates online quantile tracking as a feedback control system, utilizing proportional-integral-derivative control to achieve robust, long-run coverage guarantees with minimal interval oscillation.
- **Autocorrelated Multi-step Conformal Prediction (AcMCP)** [WH24]: An asymmetric, multi-step online calibration method that models the lag- $(h - 1)$ autocorrelation structure inherent in multi-step forecast errors using a moving average and ordinary least squares scorecaster, optimizing interval width and coverage tracking.

Adaptation to Multi-step Forecasting Because Time Series Foundation Models perform direct multi-step forecasting—outputting a sequence of future predictions over a horizon H (i.e., $\hat{y}_{t+1}, \dots, \hat{y}_{t+H}$) rather than single-step predictions—traditional one-step conformal prediction algorithms (such as ACI, PID, or Split CP) must be adapted. We employ a *step-independent marginal calibration* strategy. For a forecasting horizon H , we instantiate H independent conformal calibrators, each dedicated to a specific look-ahead step $h \in \{1, \dots, H\}$.

During calibration, nonconformity scores are computed and tracked separately for each horizon step. For these adapted methods, the control variables and interval adjustments are updated independently for each step h , based solely on the historical coverage performance of that specific horizon. This step-wise marginal calibration targets nominal coverage pointwise across all future horizons and naturally accommodates the characteristic increase in predictive uncertainty over longer look-ahead steps.

Importantly, **AcMCP** is inherently a multi-step forecasting calibration method by design. Unlike the adapted methods which treat horizons as independent calibration tasks, AcMCP recursively models the lag- $(h - 1)$ autocorrelation structure across future steps using a cross-horizon error forecasting model. Consequently, AcMCP does not require post-hoc multi-step adaptation in the same way as other methods, as multi-step cross-horizon dependencies are natively integrated into its error tracking equations.

Nonconformity Scores Rather than relying solely on standard residuals, we consider the complete spectrum of eleven physical, scaled, stateful, and quantile-grid nonconformity scores supported by our conformal evaluation pipeline:

- **Standard Absolute Residuals** (abs): The absolute point-forecast error, $s_t = |y_t - \hat{y}_t|$. This serves as the standard symmetric baseline, inverting to $[\hat{y}_t - q, \hat{y}_t + q]$.
- **Squared Residuals** (squared) [LGR⁺18]: The squared point-forecast error, $s_t = (y_t - \hat{y}_t)^2$. It penalizes outliers quadratically, inverting to the symmetric interval $[\hat{y}_t - \sqrt{\max(q, 0)}, \hat{y}_t + \sqrt{\max(q, 0)}]$.
- **Signed Residuals** (signed) [LGR⁺18]: The signed point-forecast error, $s_t = y_t - \hat{y}_t$. This score forces asymmetric calibration by constructing independent lower and upper bounds, inverting to $[\hat{y}_t - q_{lo}, \hat{y}_t + q_{up}]$.

- **IQR-Scaled Residuals** (`iqr_scaled`) [LGR⁺18]: A locally adaptive score where the absolute residual is normalized by the model's native quantile interval width, $s_t = |y_t - \hat{y}_t| / \max(q_{high,t} - q_{low,t}, \epsilon)$. It enables prediction intervals to dynamically expand or shrink according to the foundation model's in-context uncertainty, inverting to $[\hat{y}_t - q(q_{high,t} - q_{low,t}), \hat{y}_t + q(q_{high,t} - q_{low,t})]$.
- **MAD-Scaled Residuals** (`mad_scaled`) [LGR⁺18]: Normalizes the residual by a robust estimate of local residual scale based on the Median Absolute Deviation (MAD), $s_t = |y_t - \hat{y}_t| / \max(\rho_{mad,h}, \epsilon)$, where $\rho_{mad,h} = 1.4826 \times \text{MAD}(\{y_i - \hat{y}_i\}_{i \in \text{cal}})$ is computed across the calibration set for each horizon step h .
- **Conformalized Quantile Regression** (`cqr`) [RPC19]: The canonical CQR score, $s_t = \max(q_{low,t} - y_t, y_t - q_{high,t})$. It directly calibrates the native lower and upper boundaries of the model's quantiles, inverting to $[q_{low,t} - q, q_{high,t} + q]$.
- **Scaled CQR** (`scaled_cqr`) [RPC19]: Extends CQR by scaling the lower and upper violations separately using their respective robust calibration tail scales: $s_t = \max((q_{low,t} - y_t)/\sigma_{lo,h}, (y_t - q_{high,t})/\sigma_{hi,h})$, inverting to $[q_{low,t} - q\sigma_{lo,h}, q_{high,t} + q\sigma_{hi,h}]$.
- **Pinball-CRPS Like** (`distributional`) [GR07, ZN18]: The excess mean-pinball loss score evaluated over all K predicted quantile levels: $s_t = S(y_t, \hat{F}_t) - \min_z S(z, \hat{F}_t)$, where $S(y, \hat{F}_t) = \frac{1}{K} \sum_{k=1}^K \rho_{\tau_k}(y - q_{\tau_k,t})$ is the mean pinball loss. This uses the entire predicted quantile grid to calibrate a distributional-native interval, inverted through convex root-finding over the continuous pinball objective function.
- **CDF Tail** (`cdf_tail`): A distribution-native score measuring the central-band probability integral transform (PIT) violation: $s_t = \max(\alpha/2 - \hat{F}_t(y_t), \hat{F}_t(y_t) - (1 - \alpha/2), 0)$, where the cumulative distribution function $\hat{F}_t(y)$ is estimated by monotonicized piecewise-linear interpolation over the predicted quantile grid. Inversion is solved directly on the probability scale as $[Q_t(\max(0, \alpha/2 - q)), Q_t(\min(1, 1 - \alpha/2 + q))]$.
- **Log-Scaled Residuals** (`log`) [HA18]: Formulated for multiplicative-error regimes, $s_t = |\log(y_t + \epsilon) - \log(\hat{y}_t + \epsilon)|$, inverting to $[\exp(\log(\hat{y}_t + \epsilon) - q) - \epsilon, \exp(\log(\hat{y}_t + \epsilon) + q) - \epsilon]$. To prevent silent numerical errors, a domain preflight fallback automatically reverts to the abs score if nonpositive targets or predictions are detected on the calibration set.
- **Stateful Difference Residuals** (`diff`) [CWZ21]: A stateful, online nonconformity score that tracks the changes in sequential errors rather than their raw magnitude, $s_t = |e_t - e_{t-1}|$, where $e_t = y_t - \hat{y}_t$ represents the point-forecast residual. The resulting prediction interval is centered around the model's prediction shifted by the previous step's observed error: $[\hat{y}_t + e_{t-1} - q, \hat{y}_t + e_{t-1} + q]$.

3.4 Experimental Variables and Controls

To establish a rigorous, causal understanding of conformal calibration performance on pre-trained Time Series Foundation Models, we systematically control for and isolate the following experimental variables:

1. **Conformal Prediction Method and Family:** We categorize our eight evaluated conformal methods into three structural families: (a) *Static split methods* (SplitCP, WeightedCP) which establish global thresholds; (b) *Adaptive online methods* (ACI, AgACI, DtACI, Conformal PID) which sequentially adjust thresholds in response to dynamic coverage errors; and (c) *Time-series specific multi-step formulations* (AcMCP) which directly model cross-horizon error dependencies.
2. **Nonconformity Score Formulation:** We isolate the effect of the uncertainty heuristic by evaluating eleven different nonconformity scores, spanning point-residual scales (abs, squared, signed, log, diff), adaptive boundary scaling (iqr_scaled, mad_scaled, scaled_cqr), and complete quantile-grid probability models (distributional, cdf_tail).
3. **Model Architecture and Inductive Biases:** While pre-training datasets and scaling techniques differ across the evaluated foundation models, we evaluate across three fundamentally distinct architectural classes: decoder-only transformers (TimesFM), tokenized encoder-only transformers (Chronos-2), LSTM-based state-trackers (TiRex), and continuous state-space models (FlowState). This allows us to observe how underlying inductive biases influence native in-context calibration.
4. **Task Properties (Horizon, Domain, and Frequency):** To evaluate the zero-shot generalization envelope of the calibrated forecasts, we run our experiments on diverse domains in the FEV benchmark. We control for target forecast horizons ranging from short-range ($H = 5$) to ultra-long-range ($H = 288$) steps, as well as multiple sampling frequencies (annual, daily, hourly, and weekly), and varying context lengths.

3.5 Representative Configuration Selection

Due to the vast hyperparameter space spanning the cross-product of Time Series Foundation Models, conformal prediction methods, and nonconformity score types, a full exhaustive search across all possible combinations yields 398 unique configurations. To present a clear, structured, and interpretable synthesis of results in the main text (Tables 1 and 2), we select a highly representative subset of nine configurations. This selection systematically spans all major taxonomic families of conformal inference, covers the primary nonconformity score types, and includes both optimal calibrators and key pathological failures. The rigorous selection criteria and a verification analysis confirming that this subset is statistically representative of the complete 398-configuration search space are detailed in Appendix A.

3.6 Evaluation Metrics

Native quantile outputs and post-hoc CP methods are evaluated using a set of standard metrics that assess both the reliability (calibration) and informativeness (efficiency) of the generated prediction intervals $[L_{t+h}, U_{t+h}]$. For a nominal target miscoverage level $\alpha \in (0, 1)$ and forecasting horizon H , we compute the following metrics:

- **Marginal Coverage:** The empirical probability that the true target value y_{t+h} falls within the prediction interval at a specific horizon step $h \in \{1, \dots, H\}$. It is com-

puted over T test windows as:

$$\text{Marginal Coverage}(h) = \frac{1}{T} \sum_{t=1}^T \mathbb{I}(y_{t+h} \in [L_{t+h}, U_{t+h}]) \quad (1)$$

where $\mathbb{I}(\cdot)$ is the indicator function. The nominal target coverage is $1 - \alpha = 0.80$.

- **Interval Width:** Measures the efficiency of the conformalized intervals. A narrower interval indicates a more informative forecast, provided coverage guarantees are met. The physical width at horizon h is:

$$\text{Width}_t(h) = U_{t+h} - L_{t+h} \quad (2)$$

To aggregate across heterogeneous tasks with varying physical units, we report the *Scaled Width*, which normalizes the average width of a conformal method by the average width of the model's own native quantile baseline on that specific task:

$$\text{Scaled Width} = \frac{\frac{1}{T} \sum_{t=1}^T \text{Width}_t^{\text{conformal}}(h)}{\frac{1}{T} \sum_{t=1}^T \text{Width}_t^{\text{native}}(h)} \quad (3)$$

- **Winkler Score:** Jointly evaluates coverage and width, heavily penalizing miscoverage outliers. It is defined at look-ahead step h as:

$$W_t(h) = (U_{t+h} - L_{t+h}) + \frac{2}{\alpha} \max(L_{t+h} - y_{t+h}, y_{t+h} - U_{t+h}, 0) \quad (4)$$

Like widths, we report the *Scaled Winkler Score* normalized relative to the model's native quantile Winkler score.

- **Task-Normalized Width:** While Scaled Width measures the relative cost of CP calibration within a model, it does not allow direct comparison of interval efficiency *across* models. To enable this, we introduce a task-normalized width: for each task j , all raw interval widths are divided by the mean native width across all four models on that task:

$$\text{Normalized Width}_{t,j}(h) = \frac{\text{Width}_{t,j}(h)}{\frac{1}{4} \sum_{m \in \mathcal{M}} \bar{w}_j^{\text{native},m}} \quad (5)$$

where $\bar{w}_j^{\text{native},m}$ is the mean native width of model m on task j , and $\mathcal{M} = \{\text{Chronos-2, FlowState, Ti}$

Other metrics such as Continuous Ranked Probability Score (CRPS) or Quantile Loss are valuable for evaluating full probabilistic predictions, but are not directly applicable here as conformal prediction specifically calibrates a single target interval at a given level α .

4 Results and Discussion

We evaluate the calibration performance of the four pre-trained Time Series Foundation Models (TSFMs) and their post-hoc conformal calibration variants on the FEV-bench mini benchmark. All evaluations focus on a nominal target miscoverage rate of $\alpha = 0.2$,

corresponding to an 80% target prediction interval coverage ($1 - \alpha = 0.80$). We specifically choose $\alpha = 0.2$ because all foundation models in our study, with the sole exception of Chronos-2, natively only output prediction quantiles within the $[0.1, 0.9]$ interval (corresponding to a 10% to 90% quantile range). Consequently, the 80% prediction interval represents the maximum natively available confidence interval width that can be extracted out-of-the-box from these pre-trained architectures without making restrictive assumptions or relying on post-hoc extrapolation. Our empirical findings are summarized in Table 1 and Table 2, and visualized in Figure 2.

Table 1: Calibration results (coverage, scaled interval width, and scaled Winkler score) for Chronos-2 and FlowState on FEV-bench mini ($\alpha = 0.2$, Nominal Coverage $1 - \alpha = 0.80$). Widths and Winkler scores are scaled relative to the native model’s baseline. Bold indicates the best calibration method per column (closest to 0.80 for coverage, lowest for width/Winkler), and underline indicates the second best (excluding native baseline and failure cases).

Calibration Method	Coverage $\uparrow$		Scaled Width $\downarrow$		Scaled Winkler $\downarrow$	
	Task-Weighted	Series-Weighted	Task-Weighted	Series-Weighted	Task-Weighted	Series-Weighted
Chronos-2						
Native Quantiles	0.746	0.735	1.000	1.000	1.000	1.000
DtACI + CDF-Tail	0.840	0.834	1.273	1.264	1.028	1.032
ACI + CDF-Tail	0.831	<u>0.831</u>	<u>1.238</u>	1.251	<u>1.020</u>	<u>1.029</u>
DtACI + Distributional	0.833	0.849	1.402	1.836	1.158	1.272
WeightedCP + Distributional	0.820	0.881	1.254	1.650	1.127	1.139
SplitCP + CDF-Tail	<u>0.821</u>	0.805	1.192	1.170	1.011	1.023
AcMCP + CQR	<u>0.686</u>	0.634	1.573	<u>1.188</u>	1.549	1.389
TrailingWindow + Squared (Failure)	0.359	0.101	0.353	0.114	1.626	1.899
ACI + IQR-Scaled (Failure)	0.819	0.739	1.95×10^{289}	1.57×10^{290}	9.69×10^{288}	7.81×10^{289}
FlowState						
Native Quantiles	0.605	0.685	1.000	1.000	1.000	1.000
DtACI + CDF-Tail	0.720	0.792	1.298	1.262	<u>0.956</u>	<u>1.020</u>
ACI + CDF-Tail	0.720	<u>0.792</u>	<u>1.296</u>	<u>1.261</u>	0.956	1.020
DtACI + Distributional	<u>0.820</u>	0.835	1.945	2.059	1.051	1.272
WeightedCP + Distributional	0.802	0.861	1.727	1.842	1.018	1.142
SplitCP + CDF-Tail	0.709	0.771	1.274	1.205	0.958	1.017
AcMCP + CQR	0.683	0.615	1.774	1.353	1.189	1.395
TrailingWindow + Squared (Failure)	0.350	0.096	0.470	0.142	1.503	2.026
ACI + IQR-Scaled (Failure)	0.814	0.740	8.63×10^{290}	6.96×10^{291}	3.81×10^{290}	3.07×10^{291}

4.1 Coverage of Base FMs vs. CP Calibration

Our analysis reveals that native, out-of-the-box TSFM quantiles are consistently poorly calibrated and typically underestimate uncertainty in zero-shot environments. For example, FlowState exhibits a severe native coverage gap, reaching only 60.5% task-weighted coverage against the target 80%. Chronos-2 (74.6%) and TimesFM (76.6%) also fall short of the nominal target.

Applying conformal prediction (CP) methods significantly improves empirical coverage, closing the calibration gap. Most notably, CP configurations utilizing the `cdf_tail` and `distributional` nonconformity scores consistently provide the best empirical coverage, bringing it much closer to the nominal target without excessive overcoverage. For instance, on FlowState, **DtACI + Distributional** elevates coverage from 60.5% to 82.0% (task-weighted), while **WeightedCP + Distributional** achieves 80.2%, both keeping scaled interval widths modest (under $2.0\times$ the native width). For TimesFM, **SplitCP + CDF-Tail** improves coverage to 83.1%, while **DtACI + CDF-Tail** achieves 85.0%.

Table 2: Calibration results (coverage, scaled interval width, and scaled Winkler score) for TimesFM and TiRex on FEV-bench mini ($\alpha = 0.2$, Nominal Coverage $1 - \alpha = 0.80$). Widths and Winkler scores are scaled relative to the native model's baseline. Bold indicates the best calibration method per column (closest to 0.80 for coverage, lowest for width/Winkler), and underline indicates the second best (excluding native baseline and failure cases).

Calibration Method	Coverage $\uparrow$		Scaled Width $\downarrow$		Scaled Winkler $\downarrow$	
	Task-Weighted	Series-Weighted	Task-Weighted	Series-Weighted	Task-Weighted	Series-Weighted
TimesFM						
Native Quantiles	0.766	0.756	1.000	1.000	1.000	1.000
DtACI + CDF-Tail	0.850	0.848	1.280	1.267	1.032	1.049
ACI + CDF-Tail	0.838	0.844	<u>1.235</u>	1.251	<u>1.020</u>	<u>1.045</u>
DtACI + Distributional	0.833	<u>0.835</u>	1.407	1.845	1.105	1.303
WeightedCP + Distributional	0.822	0.875	1.255	1.647	1.072	1.163
SplitCP + CDF-Tail	<u>0.831</u>	0.826	1.214	1.188	1.016	1.034
AcMCP + CQR	0.708	0.650	1.327	<u>1.248</u>	1.264	1.377
TrailingWindow + Squared (Failure)	0.371	0.140	0.347	0.113	1.580	1.989
ACI + IQR-Scaled (Failure)	0.826	0.731	3.23×10^{290}	2.60×10^{291}	1.58×10^{290}	1.27×10^{291}
TiRex						
Native Quantiles	0.783	0.802	1.000	1.000	1.000	1.000
DtACI + CDF-Tail	0.860	0.876	1.262	1.225	1.034	1.056
ACI + CDF-Tail	0.846	0.872	1.210	<u>1.202</u>	<u>1.020</u>	<u>1.049</u>
DtACI + Distributional	<u>0.828</u>	0.843	1.319	1.510	1.117	1.236
WeightedCP + Distributional	0.810	0.868	<u>1.182</u>	1.372	1.089	1.112
SplitCP + CDF-Tail	0.839	<u>0.861</u>	1.167	1.151	1.010	1.039
AcMCP + CQR	0.695	0.647	3.632	1.391	2.710	1.539
TrailingWindow + Squared (Failure)	0.358	0.095	0.339	0.107	1.609	1.904
ACI + IQR-Scaled (Failure)	0.814	0.728	5.40×10^{217}	4.35×10^{218}	3.30×10^{217}	2.66×10^{218}

The success of the `cdf_tail` and `distributional` scores stems from their mathematical formulation: rather than relying on point-forecast residuals (like `abs`) or only two boundary quantiles (like `cqr`), they utilize the *entire predicted quantile grid* output by the foundation models. By interpolating the cumulative distribution function (CDF) or computing the cumulative pinball loss, they calibrate tail probabilities directly, enabling robust quantile/probability-band inversions that naturally respect distribution limits and adapt dynamically without bloating interval widths.

4.2 TiRex: Robust Native Quantiles

A key standout in our study is **TiRex**. Unlike the transformer and state-space architectures, TiRex’s native quantiles are remarkably robust, achieving 78.3% task-weighted and 80.2% series-weighted coverage out-of-the-box. This represents remarkably strong native calibration.

Consequently, post-hoc conformal prediction methods do not really improve upon TiRex’s native quantiles. In fact, applying adaptive methods like DtACI or ACI to TiRex slightly over-corrects, shifting coverage to 84%–86% while unnecessarily widening the intervals (21% to 26% wider). We hypothesize that this remarkable native calibration may be partially driven by TiRex’s underlying xLSTM recurrent state-tracking architecture. The strict recurrent state-updating of xLSTM might excel at retaining local history and variance in-context, leading to highly stable and precise quantile estimates that generalize robustly in zero-shot settings compared to attention-based or continuous state-space designs. However, other potential drivers—such as specific optimization hyper-parameters or the alignment of pre-training corpora with the target evaluation domains—cannot be ruled out, and further systematic ablation studies are needed to isolate this architectural

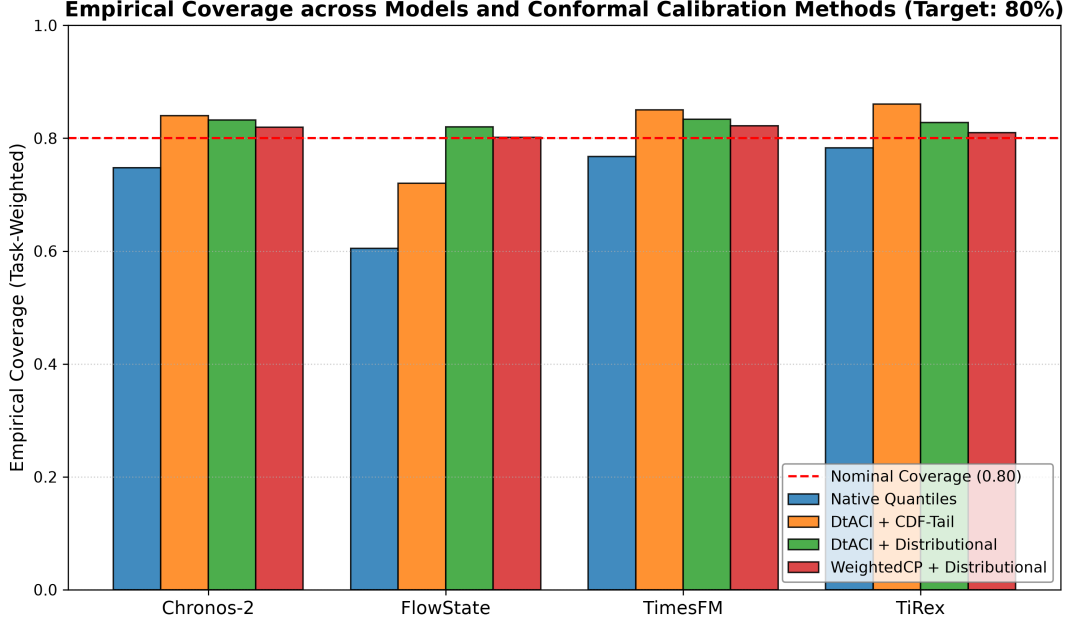


Figure 2: Empirical task-weighted coverage of native quantiles and selected conformal calibration configurations across all four evaluation models on FEV-bench mini at $\alpha = 0.20$ (nominal target $1 - \alpha = 0.80$). The red dashed line denotes the target coverage.

effect.

4.3 Cross-Model Interval Width Comparison

Tables 1 and 2 report interval widths scaled relative to each model's own native quantile baseline, which reveals the relative cost of post-hoc calibration within each model but does not allow direct comparison of interval efficiency *across* models. To enable such comparison, we introduce a *task-normalized* width metric: for each task, all interval widths (native and CP-calibrated) are divided by the mean native width across all four models on that task. This shared denominator places all models on a common scale per task, and we aggregate via task-weighted averaging. Results are presented in Table 3.

Table 3: Cross-model comparison of coverage and interval width efficiency on FEV-bench mini ($\alpha = 0.2$). Widths are task-normalized: for each task, all widths are divided by the mean native width across the four models, enabling direct cross-model comparison. Bold indicates the best model per configuration (closest to 0.80 for coverage, narrowest for width). Excludes failure cases.

Calibration Method	Coverage $\uparrow$				Normalized Width $\downarrow$			
	Chronos-2	FlowState	TimesFM	TiRex	Chronos-2	FlowState	TimesFM	TiRex
Native Quantiles	0.747	0.605	0.766	0.783	1.062	0.773	1.036	1.129
SplitCP + CDF-Tail	0.825	0.709	0.834	0.844	1.284	0.983	1.276	1.336
ACI + CDF-Tail	0.834	0.720	0.841	0.850	1.331	1.000	1.296	1.381
DtACI + CDF-Tail	0.844	0.721	0.853	0.864	1.368	1.002	1.342	1.441
WeightedCP + Distributional	0.825	0.807	0.826	0.813	1.371	1.347	1.320	1.365
DtACI + Distributional	0.841	0.828	0.842	0.836	1.546	1.538	1.502	1.561

The cross-model comparison reveals a clear coverage–width tradeoff. FlowState pro-

duces the narrowest intervals across all configurations—its native quantiles are only $0.773\times$ the cross-model mean—but this comes at the cost of the worst native coverage (60.5%), indicating its quantile estimates are systematically too tight. Even after CP correction with `cdf_tail` scores, FlowState’s intervals remain the narrowest ($0.98\text{--}1.00\times$ the cross-model mean) but its coverage still lags behind the other models (70.9%–72.1%). Only the more aggressive `distributional` score family successfully closes FlowState’s coverage gap to the 80% target, but at the expense of roughly doubling its native interval width.

Conversely, TiRex produces the widest native intervals ($1.129\times$ the cross-model mean) but achieves the best native coverage (78.3%). This suggests TiRex’s xLSTM architecture trades interval tightness for calibration accuracy—a favorable tradeoff in practice. Chronos-2 and TimesFM occupy the middle ground on both axes. Notably, the relative cost of CP calibration (the width increase factor from native to CP) is remarkably consistent across Chronos-2, TimesFM, and TiRex for `cdf_tail` scores ($\approx 1.2\text{--}1.3\times$ their own native width), while FlowState pays a proportionally larger penalty ($\approx 1.3\times$ for `cdf_tail`, $\approx 1.7\text{--}1.9\times$ for `distributional`) because its native intervals are so severely miscalibrated.

4.4 Pretraining Insights on `proenfo_gfc14`

The evaluation suite contains the `proenfo_gfc14` task, which originates from the Global Energy Forecasting Competition 2014 (GEFCom 2014) wind, solar, or load forecasting challenges. Crucially, both **TimesFM** and **FlowState** were pretrained on this dataset during their respective foundation-level training phases, while Chronos-2 and TiRex have not seen this task. This allows us to observe how foundation-level pretraining affects zero-shot (or rather, in-distribution) probabilistic calibration.

For TimesFM, pretraining appears to translate to strong in-distribution calibration natively. At nominal 80% coverage ($\alpha = 0.2$), TimesFM’s native quantiles achieve an empirical coverage of **exactly 0.800** (80.0%) with a highly efficient average interval width of 169.67 and a Winkler score of 231.45. This is in stark contrast to Chronos-2 and TiRex, which have not been trained on this task and exhibit native undercoverage (72.1% and 75.3%, respectively) alongside significantly wider intervals (376.33 and 446.84, respectively). This suggests that pretraining may enable the model’s internal representations to effectively capture the underlying data-generating process, leading to high statistical efficiency (tighter intervals that still satisfy the coverage constraint).

Conversely, FlowState presents a highly contrasting insight. Despite being pretrained on `proenfo_gfc14`, FlowState’s native quantiles suffer from severe undercoverage, reaching only 0.531 (53.1%) empirical coverage due to excessively over-confident and narrow native intervals (average width of 97.43). This demonstrates that pretraining on a dataset does not guarantee valid native quantile calibration if the underlying architecture or training objectives bias the model toward over-confidence. For FlowState, post-hoc conformal calibration appears crucial: applying conformal methods like **ACI + IQR-Scaled** successfully expands the interval width to 150.96, boosting empirical coverage to 0.720 and reducing the Winkler score from 374.09 to 330.63.

Furthermore, analyzing this task at $\alpha = 0.1$ (nominal 90% coverage) highlights a key technical benefit of conformal prediction. Because TimesFM, TiRex, and FlowState natively only output quantiles up to 0.9 and down to 0.1, their native prediction interval

is structurally limited to 80% coverage, yielding identical native results at both $\alpha = 0.1$ and $\alpha = 0.2$. Conformal prediction successfully overcomes this architectural limitation by scaling these core quantiles dynamically: at $\alpha = 0.1$, **TrailingWindow + CQR** and **WeightedCP + Distributional** successfully calibrate TimesFM to **91.0%** and **92.4%** empirical coverage, respectively, bypassing the base model’s static quantile boundaries.

4.5 Computational Complexity Takeaways

From a practical deployment standpoint, we analyze the execution times of both the foundation model inference and the post-hoc calibration methods (fully detailed in Appendix B). In brief, Chronos-2 is the most computationally efficient model (requiring only 262.55 seconds of cumulative inference), while TimesFM is the slowest (requiring 2,831.97 seconds). Regarding calibration, static split methods (SplitCP, TrailingWindow) and simpler adaptive updates (ACI, PID, AcMCP) add negligible overhead (approximately 2.1–2.5 hours across the massive high-dimensional search space). However, advanced adaptive methods incur severe bottlenecks: DtACI is the slowest, requiring 80,981.99 seconds (≈ 22.49 hours) due to sequential dual-feedback tracking. This highlights a clear trade-off between statistical adaptivity and computational scalability.

4.6 The Surprising Competitiveness of Static SplitCP

In time series forecasting—where data is characteristically non-stationary—strictly static conformal prediction methods like SplitCP are generally expected to perform poorly because they do not adapt to distribution shifts or volatility changes over time. However, our empirical results in Table 1 and Table 2 show that **SplitCP + CDF-Tail** is remarkably competitive, frequently outperforming complex adaptive online methods (e.g., achieving the best series-weighted coverage of 80.5% and 82.6% on Chronos-2 and TimesFM, respectively, while maintaining the narrowest interval widths). We identify three key factors explaining this behavior:

1. **Short Evaluation Sequences:** Across the FEV-bench mini tasks, the test evaluation set lengths are very short, typically ranging between 5 and 20 windows (often exactly 20). Over such short periods, significant non-stationary drift has little to no opportunity to manifest, meaning the exchangeability assumption between the calibration and test sets holds exceptionally well.
2. **Variance Inflation in Online Updates:** Adaptive online methods (such as ACI or DtACI) update their adjustment threshold sequentially at each step based on local feedback. Over short test horizons, this feedback loop has high variance and can over-react to random local noise or individual outliers. This introduces variance inflation (wider intervals and higher Winkler scores) without improving coverage. In contrast, SplitCP’s static threshold acts as a stable estimator.
3. **Synergy of Distributional Scores with TSFM Adaptability:** SplitCP is paired with the `cdf_tail` score, which leverages the full predicted quantile grid. Because the base TSFMs (e.g., Chronos-2, TimesFM) are highly robust and already capture time-varying patterns, seasonality, and local volatility in-context, the residuals of

the distributional nonconformity scores are clean and stationary. A static global correction is therefore sufficient.

4.7 Task-Weighted vs. Series-Weighted Evaluation

An essential methodological insight from our study is the stark contrast between task-weighted and series-weighted aggregation.

- **Series-Weighted Evaluation** assigns equal weight to every individual time-series in the meta-dataset. As a result, the aggregated metrics are heavily dominated by a few very large datasets (such as `rossmann` with 11,150 series or `boomlet` with thousands). Performance on smaller datasets with only a few series is effectively silenced.
- **Task-Weighted Evaluation** calculates the performance metric within each dataset (task) first, and then computes an unweighted average across all tasks. This treats every dataset—representing distinct domains, sampling frequencies, and forecasting horizons—equally.

We observe that series-weighted coverage is often higher than task-weighted coverage (e.g., `FlowState` native coverage jumps from 60.5% task-weighted to 68.5% series-weighted). This indicates that the base models perform better on the massive, standard datasets that dominate series-weighted averages, but struggle to generalize to the smaller, more challenging niche datasets. Task-weighted evaluation provides a much more honest and rigorous representation of zero-shot model generalization across diverse, out-of-distribution domains.

4.8 Failure Cases of Conformal Calibration

While conformal prediction offers mathematical coverage guarantees, we identify two major failure modes when integrated with pre-trained TSFMs:

Catastrophic Width Explosion When adaptive methods like ACI are combined with locally adaptive scaling scores such as `iqr_scaled` or `mad_scaled`, we observe severe interval inflation, with task-weighted widths exploding to astronomical magnitudes (e.g., 10^{290} !). This catastrophic failure occurs when:

1. The denominator scaling factor (the model's native IQR or local MAD) approaches zero in some windows due to degenerate deterministic forecasts, leading to near-division-by-zero.
2. The feedback loop in ACI continues to aggressively increment the target quantile in response to sequential outliers or non-stationary structural shocks, causing a runaway expansion.

This highlights a major vulnerability in standard online adaptive CP and underscores the necessity of bounding adjustments or utilizing smoother update schemes like DtACI.

Severe Undercoverage with Squared Scores Utilizing the squared point residual score ($s = (y - \hat{y})^2$) consistently results in catastrophic undercoverage across all models, with task-weighted coverages collapsing to approximately 35% against the 80% target. Because squaring nonconformity residuals severely overemphasizes large error outliers, the empirical quantile threshold is skewed outward to accommodate the tail. When inverted, this shrinks the normal-range prediction intervals excessively close to the median for typical, non-outlier test windows, leaving them severely unprotected.

4.9 Analysis by Task Properties

Analyzing the calibration dynamics by task characteristics highlights specific regimes of vulnerability:

Number of Calibration Windows When the number of calibration windows is extremely limited (e.g., $N_{\text{cal}} = 2$ or 1, where standard static conformal coverage like **SplitCP** collapses to 56.3% and 41.8%, respectively), post-hoc CP becomes absolutely critical. Under these data-constrained regimes, expert-ensemble or adaptive methods such as **AgACI** or **WeightedCP** combined with **distributional** scores are highly robust, recovering the target 80% coverage effectively (achieving 77.6% and 78.5% coverage for $N_{\text{cal}} = 2$, and up to 82.1% coverage for $N_{\text{cal}} = 1$, respectively).

Horizon Length For very long horizons (e.g., $H = 288$, where native coverage collapses to 57.7% due to compounding multi-step error accumulation), standard point-prediction CP methods struggle. However, **WeightedCP + Distributional** successfully models the cumulative uncertainty profile, bringing coverage back to 79.6%. This confirms that CP methods utilizing full-grid predictive distributions are indispensable for managing uncertainty over extended horizons.

For a rigorous mathematical investigation into the feedback control dynamics of online adaptive calibration, we refer the reader to Appendix D. This section provides a detailed comparative analysis of the theoretical mechanisms distinguishing ACI, AgACI, and DtACI under conformalized quantile regression (CQR), clarifying why smooth dual-tracked estimators successfully optimize the Winkler score objective.

5 Conclusion and Future Work

In this work, we conducted a comprehensive empirical evaluation of post-hoc Conformal Prediction (CP) frameworks integrated with state-of-the-art pre-trained Time Series Foundation Models (TSFMs). Our results demonstrate that native, zero-shot quantile predictions from foundation models can suffer from severe miscalibration under domain and covariate shifts. We showed that post-hoc CP methods—specifically those utilizing grid-based distributional heuristics like `cdf_tail` and `distributional`—successfully restore target coverage guarantees with minimal interval expansion. Furthermore, we uncovered key architectural trade-offs, such as the robust native calibration of recurrent (xLSTM) state-trackers like TiRex, and highlighted critical failure modes, including catastrophic width explosions under standard adaptive feedback loops.

Our findings indicate several promising directions for future research. First, while static SplitCP proved remarkably competitive on the short test sequences evaluated here, experiments over significantly larger numbers of test windows represent a crucial next step. Evaluating on longer test streams could uncover different empirical trends, particularly exposing the potential failure of static SplitCP to maintain target coverage under long-term non-stationarity, and highlighting the long-horizon benefit of adaptive online methods (such as DtACI and PID) that sequentially track distribution drift. Second, rather than deploying independent per-horizon calibration where each look-ahead step is calibrated in isolation, future work should explore methods for cross-horizon information sharing. Leveraging correlated error structures and temporal dependencies across future steps could further improve interval efficiency and joint coverage without bloating widths. Third, extending these methodologies to high-dimensional and multivariate settings represents a key milestone for industrial monitoring. Finally, developing more computationally efficient online controllers that achieve the statistical adaptivity of double-tracked methods without their sequential overhead would greatly facilitate large-scale deployment.

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A Representative Configuration Selection Criteria

Due to the vast hyperparameter space spanning the cross-product of Time Series Foundation Models, conformal prediction methods, and nonconformity score types, a full exhaustive search across all possible combinations yields 398 unique configurations. To present a clear, structured, and interpretable synthesis of results in the main text (Tables 1 and 2), we selected a highly representative subset of nine configurations. This selection is governed by three primary criteria:

1. **Systematic Representation of Method Families:** The selected configurations span all major taxonomic classes of conformal inference, including standard static baselines (SplitCP), distribution-shift-aware weighting (WeightedCP), heuristics without feedback (TrailingWindow), fully online adaptive frameworks (ACI, DtACI), and time-series-specific multi-step modeling (AcMCP).
2. **Coverage of Nonconformity Score Families:** The configurations incorporate each of the main score categories: scale-free physical absolute residuals (`abs`), outlier-magnifying squared residuals (`squared`), asymmetric point-residuals (`signed`), locally scaled residuals (`iqr_scaled`), quantile-boundary deviations (`cqr`), and grid-based distribution-native scores (`cdf_tail`, `distributional`).
3. **Delineation of Success Regimes and Failure Modes:** Rather than cherry-picking only optimal results, we select both high-performing calibrators (to show how to successfully close the calibration gap) and representative pathological failures. Specifically, we include TrailingWindow + Squared to represent undercoverage failures, and ACI + IQR-Scaled to represent the catastrophic runaway width explosion failure.

To verify that these nine configurations are representative of the larger 398-configuration search space, we perform a structured verification analysis:

- **Distributional Coverage:** The selected successful methods (SplitCP + CDF-Tail, ACI + CDF-Tail, DtACI + CDF-Tail, and WeightedCP + Distributional) consistently represent the upper-bound envelope of achievable calibration. On the other hand, AcMCP + CQR illustrates the behavior of multi-step specific formulations, while the chosen failure cases represent the extreme boundaries of undercoverage ($\approx 35\%$ coverage) and runaway width inflation ($> 10^{200}$ scaled width), ensuring the full performance distribution is spanned.
- **Score Efficiency Representation:** The selection highlights the clear superiority of grid-based scores (`cdf_tail` and `distributional`) over traditional boundary-only or absolute residuals. Pairwise comparisons (e.g., DtACI + CDF-Tail vs. DtACI + Distributional) are retained to represent how different grid inversion strategies trade interval width efficiency for marginal coverage stability.

This structured subset enables us to conduct rigorous comparative analyses across fundamentally distinct theoretical frameworks while keeping the main tables and discussion compact and accessible.

B Computational Complexity and Runtimes

To provide a complete practical assessment of the evaluated foundation models and conformal prediction methods, we report their cumulative execution times across the `fev-bench mini` test suite.

First, we measure the raw foundation model inference times (the time required to generate point forecasts and quantile distribution predictions). **Chronos-2** is the most computationally efficient, requiring only 262.55 seconds of cumulative inference time, thanks to its streamlined encoder-only token-prediction architecture. **TiRex** and **FlowState** follow at 1,521.56 seconds and 1,925.09 seconds, respectively. **TimesFM** is the slowest model, requiring 2,831.97 seconds (nearly 47 minutes) of cumulative inference, which is attributed to its multi-layer patching MLP-Transformer design and sequence-by-sequence decoding mechanism. The raw inference runtimes and relative computational overheads are summarized in Table A1.

Table A1: Foundation Model Inference Runtimes on the `fev-bench mini` test suite. Relative overhead is computed with respect to the fastest model, Chronos-2.

Foundation Model	Architectural Paradigm	Cumulative Inference Time (s)	Relative Overhead
Chronos-2	Encoder-Only Transformer	262.55	1.0×
TiRex	xLSTM Recurrent State-Tracker	1,521.56	5.8×
FlowState	Continuous State-Space Model	1,925.09	7.3×
TimesFM	Decoder-Only Patch Transformer	2,831.97	10.8×

Second, we analyze the cumulative post-hoc calibration runtimes across all evaluated tasks, models, and scores per conformal prediction method. The static and lightweight methods add negligible overhead: **TrailingWindow** (7,650.01 s), **PID** (7,692.15 s), **ACI** (7,782.25 s), **SplitCP** (8,924.94 s), and **AcMCP** (9,138.17 s) each require approximately 2.1 to 2.5 hours of cumulative runtimes across the entire high-dimensional search space. **WeightedCP** requires 16,031.60 seconds (≈ 4.45 hours) due to the overhead of computing time-decaying weighted empirical quantiles. The most complex adaptive methods exhibit significant computational bottlenecks: **AgACI** (Aggregated ACI) requires 30,452.57 seconds (≈ 8.46 hours) due to window-ensemble updates, while **DtACI** (Double-tracked ACI) is the slowest by far, requiring 80,981.99 seconds (≈ 22.49 hours) due to the sequential dual-feedback tracking and multivariate threshold updates. This highlights that while advanced adaptive methods like DtACI achieve superior coverage on non-stationary streams, they incur substantial computational overhead in large-scale deployment compared to static SplitCP. The complete calibration runtimes and overhead classifications are presented in Table A2.

C Conformal Prediction Implementation Details and Evaluation Configuration

In this appendix, we report the complete implementation details of the conformal prediction (CP) methods evaluated in our study, including the weighting scheme of **WeightedCP**, the chosen hyperparameters for all adaptive and feedback-control algorithms, and the precise task-by-task windowing configurations.

Table A2: Cumulative post-hoc conformal calibration runtimes across all evaluated tasks, models, and nonconformity scores per method. Runtimes are cumulative across the entire high-dimensional search space.

Calibration Method	Taxonomic Family	Runtime (s)	Runtime (hrs)	Computational Overhead
TrailingWindow	Rolling Quantile	7,650.01	2.13	Negligible / Real-time
Conformal PID	Feedback Control	7,692.15	2.14	Negligible / Real-time
ACI	Online Adaptive Feedback	7,782.25	2.16	Negligible / Real-time
SplitCP	Static Split Conformal	8,924.94	2.48	Negligible / Real-time
AcMCP	Autocorrelated Multi-step	9,138.17	2.54	Negligible / Real-time
WeightedCP	Weighted Conformal	16,031.60	4.45	Low / Linear-Weighting
AgACI	Expert-Ensemble Adaptive	30,452.57	8.46	Medium / Grid Updates
DtACI	Dual-Feedback Tracking	80,981.99	22.49	High / Sequential Dual-Track

C.1 WeightedCP Weighting and Quantile Estimation

The **WeightedCP** method implements rolling-window weighted conformal prediction with geometric decay weights to adapt to slow temporal distribution drift. Specifically, given a rolling window of n available calibration scores $\{s_i\}_{i=1}^n$ (where $n \leq N_{\text{cal}}$), the weights assigned to the calibration scores are computed via a geometric decay function:

$$w_i = \beta^{n-1-i}, \quad \text{for } i = 0, \dots, n-1 \quad (6)$$

where $\beta \in (0, 1]$ represents the geometric decay parameter. In all our evaluations, we set the decay parameter to the standard value of $\beta = 0.99$.

The test point at step $n+1$ is assigned a fixed weight of $w_{\text{test}} = 1.0$. The normalized probabilities p_i associated with each calibration score $i \in \{0, \dots, n-1\}$ are then defined as:

$$p_i = \frac{w_i}{\sum_{j=0}^{n-1} w_j + w_{\text{test}}} \quad (7)$$

The conformal threshold $\hat{q}$ is computed as the $(1 - \alpha_{\text{eff}})$ weighted empirical quantile of the calibration scores. Under the finite-sample conformal correction option, the target quantile probability α_{eff} is adjusted as:

$$1 - \alpha_{\text{eff}} = \min \left(\frac{\lceil (n+1)(1 - \alpha) \rceil}{n}, 1.0 \right) \quad (8)$$

Otherwise, the uncorrected target probability $1 - \alpha_{\text{eff}} = 1 - \alpha$ is used.

C.2 Chosen Methodological Hyperparameters

The hyperparameters for all post-hoc conformal prediction controllers were selected based on standard theoretical defaults from literature to ensure a fair and robust benchmark comparison:

- **SplitCP**: Strictly static baseline. It computes a global threshold over the entire active calibration set once, with no sequential updates during the test phase.
- **TrailingWindow**: Rolling window threshold estimator. It computes a standard empirical quantile over a sliding window of the most recent scores of size W , where W matches the active calibration window size.

- **ACI (Adaptive Conformal Inference)**: Updates the target miscoverage rate sequentially using simple proportional integral feedback: $\alpha_{t+1} = \alpha_t + \gamma(\alpha - \text{err}_t)$. We set the feedback step size (learning rate) to $\gamma = 0.005$ across all tasks.
- **Conformal PID (QuantileIntegrator)**: Proportional-Integral-Derivative feedback controller. We instantiate the proportional-only variant ($K_I = 0.0$), matching Conformal Quantile Control (CQC), with proportional learning rate $lr = 0.1$. The step size is dynamically scaled by the range of scores in the rolling window (`proportional_lr=True`) to ensure scale-invariance.
- **AgACI (Aggregated ACI)**: Mixes $K = 5$ independent ACI experts with log-spaced candidate step sizes $\gamma_k \in \{0.001, 0.005, 0.01, 0.05, 0.1\}$. The mixing weights are updated online using the AdaHedge FTRL rule with a numerical stability parameter $\epsilon = 10^{-3}$.
- **DtACI (Double-tracked ACI)**: Maintains an ensemble of $K = 3$ experts with candidate step-sizes $\gamma_k \in \{0.001, 0.01, 0.1\}$, expert mixing parameter $\sigma = 10^{-3}$, and static exponential learning parameter $\eta = 2.72$.
- **AcMCP**: Autocorrelated Multi-step conformal prediction. Uses a score-prediction learning rate $lr = 0.1$, training burn-in length $n_{\text{cal}} = 10$, and initializes the integrator gain K_I to the maximum calibration error.

C.3 Evaluation Windowing and Disjoint Splitting Strategy

We strictly follow the original temporal splitting and windowing methodology defined in the `fev-bench` benchmark framework (Shchur et al., 2025; arXiv:2509.26468).

Let T denote the total length of a given time series, and let H represent the prediction horizon (step size). The series is divided into a sequence of sliding windows of size H . To ensure strict temporal separation and eliminate any potential lookahead or leakage bias, the dataset is split into two disjoint sets:

1. **Test Windows**: Evaluated on exactly the same number of test windows (N_{test}) as specified in the original FEV benchmark task definition.
2. **Calibration Windows**: Constructed from a contiguous block of windows immediately preceding the test windows. The maximum possible calibration windows $N_{\text{cal_max}}$ is determined by the series length remaining after accounting for the test windows and leaving at least one window’s worth of context at the beginning:

$$N_{\text{cal_max}} = \left\lfloor \frac{T}{H} \right\rfloor - N_{\text{test}} - 1 \quad (9)$$

When a target calibration window size N_{target} (e.g., $N_{\text{target}} = 50$) is specified via the `--cal-windows` command-line argument, the active calibration set is clipped to the most recent $M = \min(N_{\text{target}}, N_{\text{cal_max}})$ windows immediately preceding the test set, discarding any earlier windows. For tasks with fewer available windows than N_{target} , the evaluation seamlessly utilizes all available $N_{\text{cal_max}}$ calibration windows (and tasks with $N_{\text{cal_max}} < N_{\text{min_cal_windows}}$ are skipped). This temporal windowing split is detailed in Table A3.

Table A3: Task-by-task windowing configurations and dataset dimensions across the `fev-bench mini` evaluation suite. Total windows represent the sum of all available calibration windows (N_{cal}) and test windows (N_{test}).

Task Name	Horizon (H)	Total Windows	Max Cal Windows	Test Windows	Num Series
ETT_15T	96	724	704	20	14
ETT_1H	168	102	82	20	14
M_DENSE_1D	28	25	15	10	30
bizitobs_l2c_5T	288	110	90	20	7
boomlet_1282	60	272	252	20	35
boomlet_1676	96	107	87	20	100
boomlet_619	60	272	252	20	52
epf_np	24	2183	2163	20	1
hospital_admissions_1D	28	60	40	20	8
hospital_admissions_1W	13	17	1	16	8
jena_weather_1H	24	365	345	20	21
proenfo_gfc14	168	103	83	20	1
rohlik_orders_1D	61	11	6	5	7
rossmann_1D	48	18	8	10	1115
rossmann_1W	13	9	1	8	1115
solar_with_weather_1H	24	2067	2047	20	1
uci_air_quality_1H	168	54	34	20	4
uk_covid_nation_1D/cumulative	28	25	5	20	12
us_consumption_1Y	5	11	1	10	31
world_co2_emissions	5	11	2	9	191

D Theoretical Comparison of Online Calibration Methods under CQR

When applying conformalized quantile regression (CQR) scores to pre-trained Time Series Foundation Models (TSFMs) such as Chronos-2, the nonconformity score at time t for a given horizon step is defined as:

$$s_t = \max(q_{\text{low},t} - y_t, y_t - q_{\text{high},t}) \quad (10)$$

where $q_{\text{low},t}$ and $q_{\text{high},t}$ represent the native lower and upper quantile predictions. Because the foundation model's native quantile outputs are already pre-calibrated to a target coverage rate of $1 - \alpha$, the distribution of s_t exhibits a high density (point mass) around 0, with a tail of positive values corresponding to rare boundary violations.

Applying adaptive calibration methods to these scores reveals stark differences in performance, particularly when evaluated using the Winkler score:

$$W_t = (U_t - L_t) + \frac{2}{\alpha} \max(L_t - y_t, y_t - U_t, 0) \quad (11)$$

where $[L_t, U_t]$ is the constructed prediction interval. The major differences between ACI, AgACI, and DtACI are detailed below.

Continuous vs. Binary Feedback Adaptive Conformal Inference (ACI) [GC21] and its aggregated variant AgACI [ZFG⁺22] rely on a binary feedback loop:

$$e_t = \mathbb{I}(s_t > q_t) \in \{0, 1\} \quad (12)$$

which updates the nominal parameters via $\alpha_{t+1} = \alpha_t + \gamma(\alpha - e_t)$. Near the spike at 0 in the CQR score distribution, this binary feedback creates severe *on-off oscillations*. Small

noise fluctuations induce sequential violations ($e_t = 1$) followed by consecutive coverages ($e_t = 0$). This causes the intervals to alternate between being overly wide (inflating the width term of the Winkler score) and overly narrow (incurring the heavy $\frac{2}{\alpha}$ miscoverage penalty), resulting in a worse Winkler score than the native baseline.

Conversely, DtACI [GC24] utilizes the continuous empirical error rate $\beta_t \in [0, 1]$ of the new score relative to the rolling window of scores:

$$\beta_t = \frac{1}{n} \sum_{i=1}^n \mathbb{I}(s_{t-i} > s_t) \quad (13)$$

By replacing the binary feedback with a smooth quantile location, DtACI eliminates high-frequency boundary oscillations, leading to stable interval widths.

Parameter Averaging vs. Randomized Sampling AgACI aggregates a grid of candidate step-sizes by taking a linear convex combination of the experts' nominal levels:

$$\alpha_t = \sum_{k=1}^K p_t^k \alpha_t^k \quad (14)$$

Because the mapping from the nominal level α_t to the empirical quantile $Q(1 - \alpha_t)$ is highly non-linear, parameter averaging is mathematically sub-optimal. It often yields biased quantiles that over-cover without optimizing interval efficiency.

DtACI solves this by randomly sampling the expert at each step according to the probability vector p_t . This guarantees that the aggregated sequence inherits the regret bounds of the best performing expert.

Equivalence to Pinball Loss Minimization The Winkler score is mathematically equivalent to the pinball loss $\rho_\alpha(\cdot)$ up to a scaling factor. ACI and AgACI are formulated as reactive controllers that do not optimize a specific loss. DtACI, however, is explicitly framed within the context of online convex optimization, updating its expert weights by minimizing the pinball loss of the empirical error rates:

$$\ell(\beta_t, \alpha_{\text{exp}}) = \rho_\alpha(\beta_t - \alpha_{\text{exp}}) \quad (15)$$

Because DtACI directly minimizes the pinball loss of the quantiles, it is theoretically guaranteed to minimize the objective measured by the Winkler score.

E Complete Grid Search Results

In this appendix, we present the complete empirical results for all 398 combinations of base Time Series Foundation Models, conformal prediction methods, and nonconformity score types evaluated on the FEV-bench mini benchmark at a nominal miscoverage level of $\alpha = 0.2$ (target 80% coverage).

E.1 Coverage vs. Width Grid Visualization

Figure A1 displays the trade-off between task-weighted empirical coverage and task-weighted scaled interval width across all combinations. In each panel, every marker represents a unique configuration of conformal prediction method and nonconformity score. High-quality configurations cluster near the red dashed nominal coverage target of 0.80 while maintaining a small scaled width (closer to 1.0). Configurations that suffer from interval width explosion (scaled width > 100) are clipped and plotted at the far-right edge of the subplots (indicated by 'x' markers).

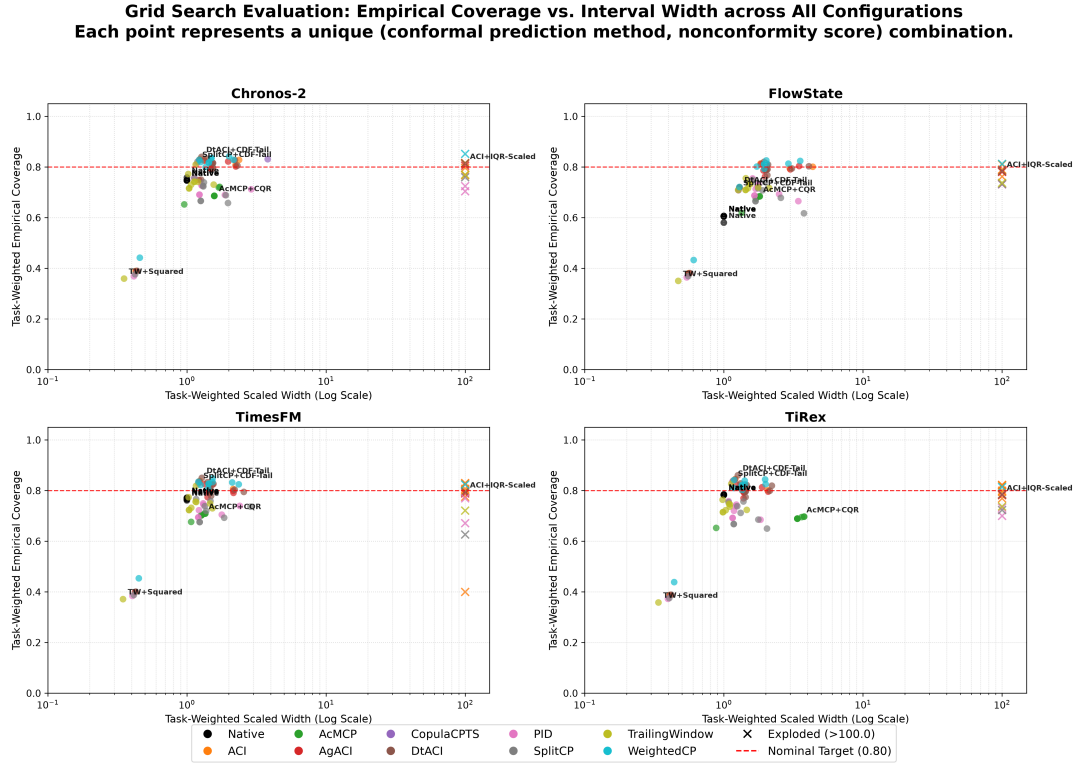


Figure A1: Task-weighted empirical coverage vs. task-weighted scaled interval width for all evaluated configurations across Chronos-2, FlowState, TimesFM, and TiRex on FEV-bench mini ($\alpha = 0.2$). The red dashed line denotes the target coverage of 80%.

E.2 Complete Results for Chronos-2

The complete evaluation results of all conformal prediction configurations for **Chronos-2** on FEV-bench mini at $\alpha = 0.2$ are presented in Table A4. Widths are scaled relative to the model's native baseline.

Table A4: All conformal prediction configurations for Chronos-2 on FEV-bench mini ($\alpha = 0.2$, Nominal Coverage $1 - \alpha = 0.80$). Columns report Task-Weighted (Task) and Series-Weighted (Series) aggregations.

Calibration Method	Coverage $\uparrow$		Scaled Width $\downarrow$	
	Task	Series	Task	Series
ACI + abs	0.768	0.754	3.23×10^{285}	3.23×10^{286}
ACI + cdf-tail	0.831	0.831	1.238	1.251
ACI + cqr	0.816	0.733	7.47×10^{213}	6.02×10^{214}
ACI + diff	0.808	0.748	2.238	2.206
ACI + distributional	0.828	0.848	1.331	1.773
ACI + iqr-scaled	0.813	0.761	1.08×10^{289}	1.04×10^{290}
ACI + log	0.828	0.804	2.372	2.890
ACI + mad-scaled	0.797	0.733	1.495	1.520
ACI + scaled-cqr	0.816	0.733	8.60×10^{288}	6.93×10^{289}
ACI + signed	0.797	0.733	7.47×10^{213}	6.02×10^{214}
ACI + squared	0.389	0.110	0.428	0.139
AcMCP + abs	0.722	0.735	1.721	1.520
AcMCP + cdf-tail	0.686	0.634	1.573	1.188
AcMCP + cqr	0.686	0.634	1.573	1.188
AcMCP + diff	0.686	0.634	1.573	1.188
AcMCP + distributional	0.686	0.634	1.573	1.188
AcMCP + iqr-scaled	0.719	0.696	1.696	1.389
AcMCP + log	0.652	0.547	0.957	0.782
AcMCP + mad-scaled	0.686	0.634	1.573	1.188
AcMCP + scaled-cqr	0.686	0.634	1.573	1.188
AcMCP + signed	0.686	0.634	1.573	1.188
AcMCP + squared	0.686	0.634	1.573	1.188
AgACI + abs	0.748	0.749	1.222	1.377
AgACI + cdf-tail	0.826	0.830	1.225	1.244
AgACI + cqr	0.813	0.731	1.508	1.478
AgACI + diff	0.802	0.745	2.248	2.194
AgACI + distributional	0.825	0.847	1.335	1.793
AgACI + iqr-scaled	0.805	0.758	1.25×10^7	1.20×10^8
AgACI + log	0.822	0.798	1.980	2.432
AgACI + mad-scaled	0.794	0.731	1.494	1.504
AgACI + scaled-cqr	0.813	0.731	1.508	1.478
AgACI + signed	0.794	0.731	1.494	1.504
AgACI + squared	0.389	0.110	0.432	0.140

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Table A4 – Continued from previous page

Calibration Method	Coverage $\uparrow$		Scaled Width $\downarrow$	
	Task	Series	Task	Series
DtACI + abs	0.750	0.757	1.264	1.423
DtACI + cdf-tail	0.840	0.834	1.273	1.264
DtACI + cqr	0.815	0.731	1.540	1.477
DtACI + diff	0.806	0.745	2.308	2.195
DtACI + distributional	0.833	0.849	1.402	1.836
DtACI + iqr-scaled	0.815	0.770	1.23×10^7	1.15×10^8
DtACI + log	0.827	0.800	2.213	2.662
DtACI + mad-scaled	0.797	0.731	1.538	1.504
DtACI + scaled-cqr	0.816	0.731	1.539	1.478
DtACI + signed	0.796	0.731	1.539	1.506
DtACI + squared	0.390	0.109	0.439	0.142
Native Quantiles	0.747	0.729	1.000	1.000
PID + abs	0.712	0.687	2.901	2.817
PID + cdf-tail	0.811	0.805	1.172	1.167
PID + cqr	0.727	0.532	1.277	0.934
PID + diff	0.704	0.514	2.92×10^{285}	2.35×10^{286}
PID + distributional	0.756	0.754	1.128	1.457
PID + iqr-scaled	0.725	0.604	1.29×10^7	1.24×10^8
PID + log	0.690	0.605	1.875	1.976
PID + mad-scaled	0.691	0.511	1.231	0.976
PID + scaled-cqr	0.727	0.532	1.278	0.936
PID + signed	0.691	0.511	1.231	0.976
PID + squared	0.369	0.103	0.416	0.134
SplitCP + abs	0.739	0.682	1.325	1.368
SplitCP + cdf-tail	0.821	0.805	1.192	1.170
SplitCP + cqr	0.726	0.480	1.307	0.869
SplitCP + diff	0.689	0.456	1.905	1.419
SplitCP + distributional	0.757	0.736	1.134	1.484
SplitCP + iqr-scaled	0.760	0.555	1.17×10^7	1.12×10^8
SplitCP + log	0.658	0.529	1.974	2.085
SplitCP + mad-scaled	0.666	0.440	1.258	0.927
SplitCP + scaled-cqr	0.723	0.478	1.312	0.868
SplitCP + signed	0.666	0.440	1.258	0.927
SplitCP + squared	0.377	0.104	0.425	0.138
TrailingWindow + abs	0.730	0.746	1.073	1.230
TrailingWindow + cdf-tail	0.809	0.805	1.156	1.149
TrailingWindow + cqr	0.742	0.683	1.132	1.137
TrailingWindow + diff	0.731	0.703	1.558	1.735
TrailingWindow + distributional	0.772	0.773	1.023	1.159
TrailingWindow + iqr-scaled	0.770	0.727	8.20×10^6	7.86×10^7
TrailingWindow + log	0.741	0.720	1.201	1.249
TrailingWindow + mad-scaled	0.716	0.688	1.037	1.150
TrailingWindow + scaled-cqr	0.742	0.683	1.131	1.136

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Table A4 – Continued from previous page

Calibration Method	Coverage $\uparrow$		Scaled Width $\downarrow$	
	Task	Series	Task	Series
TrailingWindow + signed	0.716	0.688	1.038	1.150
TrailingWindow + squared	0.359	0.101	0.353	0.114
WeightedCP + abs	0.800	0.839	1.363	1.890
WeightedCP + cdf-tail	0.829	0.840	1.226	1.269
WeightedCP + cqr	0.835	0.857	1.500	2.038
WeightedCP + diff	0.828	0.850	2.164	2.945
WeightedCP + distributional	0.820	0.881	1.254	1.650
WeightedCP + iqr-scaled	0.852	0.874	1.31×10^7	1.54×10^8
WeightedCP + log	0.842	0.880	2.009	2.721
WeightedCP + mad-scaled	0.818	0.845	1.438	2.033
WeightedCP + scaled-cqr	0.835	0.857	1.500	2.038
WeightedCP + signed	0.818	0.845	1.438	2.033
WeightedCP + squared	0.442	0.182	0.459	0.212

E.3 Complete Results for FlowState

The complete evaluation results of all conformal prediction configurations for **FlowState** on FEV-bench mini at $\alpha = 0.2$ are presented in Table A5. Widths are scaled relative to the model's native baseline.

Table A5: All conformal prediction configurations for FlowState on FEV-bench mini ($\alpha = 0.2$, Nominal Coverage $1 - \alpha = 0.80$). Columns report Task-Weighted (Task) and Series-Weighted (Series) aggregations.

Calibration Method	Coverage $\uparrow$		Scaled Width $\downarrow$	
	Task	Series	Task	Series
ACI + abs	0.762	0.724	4.63×10^{276}	3.94×10^{277}
ACI + cdf-tail	0.720	0.792	1.296	1.261
ACI + cqr	0.788	0.733	1.08×10^{286}	9.49×10^{286}
ACI + diff	0.794	0.736	2.981	2.526
ACI + distributional	0.814	0.835	1.836	1.992
ACI + iqr-scaled	0.781	0.736	4.93×10^{290}	4.60×10^{291}
ACI + log	0.801	0.799	4.359	4.820
ACI + mad-scaled	0.790	0.723	1.45×10^{217}	1.17×10^{218}
ACI + scaled-cqr	0.810	0.735	7.65×10^{289}	6.17×10^{290}
ACI + signed	0.790	0.723	1.998	1.714
ACI + squared	0.381	0.104	0.560	0.170
AcMCP + abs	0.684	0.636	1.812	1.430
AcMCP + cdf-tail	0.676	0.582	1.702	1.239
AcMCP + cqr	0.683	0.615	1.774	1.353
AcMCP + diff	0.676	0.582	1.702	1.239

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Table A5 – Continued from previous page

Calibration Method	Coverage ↑		Scaled Width ↓	
	Task	Series	Task	Series
AcMCP + distributional	0.676	0.582	1.702	1.239
AcMCP + iqr-scaled	0.684	0.636	1.812	1.430
AcMCP + log	0.619	0.558	1.345	1.186
AcMCP + mad-scaled	0.676	0.582	1.702	1.239
AcMCP + scaled-cqr	0.676	0.582	1.702	1.239
AcMCP + signed	0.676	0.582	1.702	1.239
AcMCP + squared	0.676	0.582	1.702	1.239
AgACI + abs	0.762	0.723	1.978	1.135
AgACI + cdf-tail	0.717	0.790	1.289	1.257
AgACI + cqr	0.786	0.732	1.928	1.675
AgACI + diff	0.791	0.734	3.007	2.512
AgACI + distributional	0.812	0.834	1.854	2.016
AgACI + iqr-scaled	0.781	0.735	24033.8	230257.1
AgACI + log	0.804	0.797	3.478	3.894
AgACI + mad-scaled	0.787	0.721	2.003	1.702
AgACI + scaled-cqr	0.806	0.734	2.040	1.701
AgACI + signed	0.787	0.721	2.003	1.702
AgACI + squared	0.380	0.104	0.563	0.170
DtACI + abs	0.768	0.733	2.055	1.125
DtACI + cdf-tail	0.720	0.792	1.298	1.262
DtACI + cqr	0.791	0.738	1.973	1.757
DtACI + diff	0.794	0.733	3.085	2.516
DtACI + distributional	0.820	0.835	1.945	2.059
DtACI + iqr-scaled	0.785	0.745	23566.9	216372.4
DtACI + log	0.803	0.795	4.087	4.478
DtACI + mad-scaled	0.791	0.720	2.057	1.703
DtACI + scaled-cqr	0.810	0.733	2.079	1.699
DtACI + signed	0.791	0.720	2.055	1.701
DtACI + squared	0.381	0.104	0.570	0.172
Native Quantiles	0.607	0.676	1.000	1.000
PID + abs	0.714	0.623	1.694	0.842
PID + cdf-tail	0.708	0.770	1.274	1.202
PID + cqr	0.732	0.585	1.770	0.943
PID + diff	0.693	0.503	2.496	1.668
PID + distributional	0.755	0.725	1.610	1.633
PID + iqr-scaled	0.731	0.633	12827.7	109220.2
PID + log	0.665	0.596	3.432	3.508
PID + mad-scaled	0.687	0.505	1.661	1.104
PID + scaled-cqr	0.724	0.517	1.750	1.099
PID + signed	0.687	0.505	1.661	1.104
PID + squared	0.364	0.098	0.541	0.164
SplitCP + abs	0.708	0.546	1.898	1.372
SplitCP + cdf-tail	0.709	0.771	1.274	1.205

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Table A5 – Continued from previous page

Calibration Method	Coverage $\uparrow$		Scaled Width $\downarrow$	
	Task	Series	Task	Series
SplitCP + cqr	0.741	0.533	1.972	1.234
SplitCP + diff	0.678	0.446	2.570	1.606
SplitCP + distributional	0.749	0.694	1.613	1.651
SplitCP + iqr-scaled	0.732	0.558	18735.6	174497.5
SplitCP + log	0.617	0.535	3.767	3.787
SplitCP + mad-scaled	0.665	0.431	1.690	1.051
SplitCP + scaled-cqr	0.716	0.469	1.765	1.035
SplitCP + signed	0.665	0.431	1.690	1.051
SplitCP + squared	0.369	0.099	0.555	0.170
TrailingWindow + abs	0.716	0.693	1.492	0.963
TrailingWindow + cdf-tail	0.709	0.772	1.268	1.206
TrailingWindow + cqr	0.732	0.697	1.552	1.371
TrailingWindow + diff	0.717	0.689	2.120	1.987
TrailingWindow + distributional	0.756	0.758	1.439	1.329
TrailingWindow + iqr-scaled	0.737	0.707	18082.1	166020.6
TrailingWindow + log	0.720	0.717	1.766	1.862
TrailingWindow + mad-scaled	0.709	0.676	1.435	1.302
TrailingWindow + scaled-cqr	0.734	0.690	1.521	1.327
TrailingWindow + signed	0.710	0.676	1.434	1.302
TrailingWindow + squared	0.350	0.096	0.470	0.142
WeightedCP + abs	0.793	0.821	1.953	1.449
WeightedCP + cdf-tail	0.721	0.806	1.294	1.278
WeightedCP + cqr	0.814	0.841	2.039	2.382
WeightedCP + diff	0.813	0.843	2.912	3.509
WeightedCP + distributional	0.802	0.861	1.727	1.842
WeightedCP + iqr-scaled	0.812	0.834	20701.3	235637.7
WeightedCP + log	0.824	0.882	3.541	4.400
WeightedCP + mad-scaled	0.811	0.843	1.947	2.394
WeightedCP + scaled-cqr	0.826	0.861	2.021	2.416
WeightedCP + signed	0.811	0.843	1.947	2.394
WeightedCP + squared	0.433	0.173	0.607	0.264

E.4 Complete Results for TimesFM

The complete evaluation results of all conformal prediction configurations for **TimesFM** on FEV-bench mini at $\alpha = 0.2$ are presented in Table A6. Widths are scaled relative to the model’s native baseline.

Table A6: All conformal prediction configurations for TimesFM on FEV-bench mini ($\alpha = 0.2$, Nominal Coverage $1 - \alpha = 0.80$). Columns report Task-Weighted (Task) and Series-Weighted (Series) aggregations.

Calibration Method	Coverage $\uparrow$		Scaled Width $\downarrow$	
	Task	Series	Task	Series
ACI + abs	0.775	0.726	7.82×10^{286}	6.66×10^{287}
ACI + cdf-tail	0.838	0.844	1.235	1.251
ACI + cqr	0.807	0.738	5.84×10^{286}	5.15×10^{287}
ACI + diff	0.806	0.740	2.149	2.211
ACI + distributional	0.830	0.835	7.11×10^{285}	5.73×10^{286}
ACI + iqr-scaled	0.793	0.726	1.84×10^{290}	1.72×10^{291}
ACI + log	0.797	0.722	7.38×10^7	4.55×10^8
ACI + mad-scaled	0.798	0.722	8.65×10^{289}	6.97×10^{290}
ACI + scaled-cqr	0.828	0.739	9.54×10^{288}	7.68×10^{289}
ACI + signed	0.799	0.722	8.04×10^{286}	6.48×10^{287}
ACI + squared	0.400	0.146	4.74×10^{138}	3.82×10^{139}
AcMCP + abs	0.709	0.671	1.356	1.321
AcMCP + cdf-tail	0.702	0.618	1.271	1.139
AcMCP + cqr	0.708	0.650	1.327	1.248
AcMCP + diff	0.702	0.618	1.271	1.139
AcMCP + distributional	0.702	0.618	1.271	1.139
AcMCP + iqr-scaled	0.709	0.671	1.356	1.321
AcMCP + log	0.676	0.602	1.073	1.101
AcMCP + mad-scaled	0.702	0.618	1.271	1.139
AcMCP + scaled-cqr	0.702	0.618	1.271	1.139
AcMCP + signed	0.702	0.618	1.271	1.139
AcMCP + squared	0.702	0.618	1.271	1.139
AgACI + abs	0.774	0.725	1.409	1.001
AgACI + cdf-tail	0.835	0.842	1.224	1.245
AgACI + cqr	0.805	0.738	1.433	1.493
AgACI + diff	0.799	0.738	2.152	2.199
AgACI + distributional	0.826	0.834	1.342	1.799
AgACI + iqr-scaled	0.791	0.726	2.162	1.973
AgACI + log	0.789	0.721	7.06×10^7	4.34×10^8
AgACI + mad-scaled	0.793	0.720	1.447	1.506
AgACI + scaled-cqr	0.823	0.738	1.525	1.509
AgACI + signed	0.793	0.720	1.447	1.506
AgACI + squared	0.399	0.146	0.423	0.138
DtACI + abs	0.779	0.734	1.492	0.995
DtACI + cdf-tail	0.850	0.848	1.280	1.267
DtACI + cqr	0.809	0.743	1.470	1.583
DtACI + diff	0.803	0.738	2.205	2.199
DtACI + distributional	0.833	0.835	1.407	1.845

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Table A6 – Continued from previous page

Calibration Method	Coverage $\uparrow$		Scaled Width $\downarrow$	
	Task	Series	Task	Series
DtACI + iqr-scaled	0.795	0.735	2.565	2.125
DtACI + log	0.796	0.720	7.06×10^7	4.34×10^8
DtACI + mad-scaled	0.798	0.719	1.493	1.506
DtACI + scaled-cqr	0.826	0.738	1.552	1.512
DtACI + signed	0.798	0.719	1.493	1.506
DtACI + squared	0.401	0.146	0.429	0.140
Native Quantiles	0.772	0.751	1.000	1.000
PID + abs	0.722	0.632	1.221	0.734
PID + cdf-tail	0.827	0.825	1.205	1.182
PID + cqr	0.750	0.607	1.309	0.843
PID + diff	0.705	0.512	1.782	1.492
PID + distributional	0.769	0.730	8.24×10^{16}	6.64×10^{17}
PID + iqr-scaled	0.740	0.631	2.399	0.828
PID + log	0.671	0.494	2.18×10^6	1.19×10^7
PID + mad-scaled	0.694	0.519	1.208	0.987
PID + scaled-cqr	0.744	0.545	1.316	0.997
PID + signed	0.694	0.519	1.208	0.987
PID + squared	0.384	0.141	0.406	0.132
SplitCP + abs	0.720	0.566	1.376	1.243
SplitCP + cdf-tail	0.831	0.826	1.214	1.188
SplitCP + cqr	0.759	0.568	1.477	1.116
SplitCP + diff	0.693	0.455	1.854	1.451
SplitCP + distributional	0.766	0.700	1.165	1.464
SplitCP + iqr-scaled	0.736	0.546	2.861	3.840
SplitCP + log	0.626	0.430	2.25×10^6	1.09×10^7
SplitCP + mad-scaled	0.676	0.447	1.235	0.952
SplitCP + scaled-cqr	0.738	0.508	1.345	0.955
SplitCP + signed	0.676	0.447	1.235	0.952
SplitCP + squared	0.388	0.142	0.415	0.136
TrailingWindow + abs	0.732	0.696	1.075	0.831
TrailingWindow + cdf-tail	0.819	0.816	1.163	1.147
TrailingWindow + cqr	0.755	0.703	1.167	1.230
TrailingWindow + diff	0.730	0.698	1.526	1.740
TrailingWindow + distributional	0.774	0.759	1.029	1.163
TrailingWindow + iqr-scaled	0.749	0.693	1.468	1.606
TrailingWindow + log	0.721	0.681	6.88×10^7	4.26×10^8
TrailingWindow + mad-scaled	0.723	0.677	1.036	1.148
TrailingWindow + scaled-cqr	0.756	0.695	1.154	1.185
TrailingWindow + signed	0.723	0.677	1.036	1.147
TrailingWindow + squared	0.371	0.140	0.347	0.113
WeightedCP + abs	0.810	0.830	1.421	1.236
WeightedCP + cdf-tail	0.835	0.854	1.209	1.258
WeightedCP + cqr	0.834	0.853	1.530	2.099

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Table A6 – Continued from previous page

Calibration Method	Coverage $\uparrow$		Scaled Width $\downarrow$	
	Task	Series	Task	Series
WeightedCP + diff	0.832	0.857	2.116	3.043
WeightedCP + distributional	0.822	0.875	1.255	1.647
WeightedCP + iqr-scaled	0.825	0.834	2.359	5.608
WeightedCP + log	0.824	0.892	2.65×10^6	2.36×10^7
WeightedCP + mad-scaled	0.827	0.856	1.434	2.084
WeightedCP + scaled-cqr	0.848	0.876	1.531	2.103
WeightedCP + signed	0.827	0.856	1.434	2.084
WeightedCP + squared	0.454	0.242	0.451	0.210

E.5 Complete Results for TiRex

The complete evaluation results of all conformal prediction configurations for **TiRex** on FEV-bench mini at $\alpha = 0.2$ are presented in Table A7. Widths are scaled relative to the model’s native baseline.

Table A7: All conformal prediction configurations for TiRex on FEV-bench mini ($\alpha = 0.2$, Nominal Coverage $1 - \alpha = 0.80$). Columns report Task-Weighted (Task) and Series-Weighted (Series) aggregations.

Calibration Method	Coverage $\uparrow$		Scaled Width $\downarrow$	
	Task	Series	Task	Series
ACI + abs	0.769	0.727	3.25×10^{276}	2.77×10^{277}
ACI + cdf-tail	0.846	0.872	1.210	1.202
ACI + cqr	0.799	0.729	3.78×10^{285}	3.33×10^{286}
ACI + diff	0.801	0.748	2.066	1.839
ACI + distributional	0.822	0.843	4.31×10^{286}	3.47×10^{287}
ACI + iqr-scaled	0.782	0.726	3.09×10^{217}	2.87×10^{218}
ACI + log	0.820	0.796	7.05×10^{235}	2.49×10^{235}
ACI + mad-scaled	0.795	0.726	1.392	1.252
ACI + scaled-cqr	0.822	0.732	1.32×10^{280}	1.06×10^{281}
ACI + signed	0.795	0.726	5.20×10^{285}	4.19×10^{286}
ACI + squared	0.388	0.102	0.409	0.130
AcMCP + abs	0.697	0.666	3.775	1.489
AcMCP + cdf-tail	0.689	0.617	3.377	1.246
AcMCP + cqr	0.695	0.647	3.632	1.391
AcMCP + diff	0.689	0.617	3.377	1.246
AcMCP + distributional	0.689	0.617	3.377	1.246
AcMCP + iqr-scaled	0.697	0.666	3.775	1.489
AcMCP + log	0.653	0.556	0.881	0.691
AcMCP + mad-scaled	0.689	0.617	3.377	1.246
AcMCP + scaled-cqr	0.689	0.617	3.377	1.246

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Table A7 – Continued from previous page

Calibration Method	Coverage $\uparrow$		Scaled Width $\downarrow$	
	Task	Series	Task	Series
AcMCP + signed	0.689	0.617	3.377	1.246
AcMCP + squared	0.689	0.617	3.377	1.246
AgACI + abs	0.770	0.726	1.386	0.827
AgACI + cdf-tail	0.843	0.869	1.200	1.195
AgACI + cqr	0.800	0.728	1.358	1.252
AgACI + diff	0.796	0.745	2.083	1.830
AgACI + distributional	0.821	0.840	1.256	1.475
AgACI + iqr-scaled	0.782	0.725	1.93×10^7	1.85×10^8
AgACI + log	0.813	0.791	1.885	2.124
AgACI + mad-scaled	0.793	0.724	1.396	1.245
AgACI + scaled-cqr	0.819	0.730	1.434	1.270
AgACI + signed	0.793	0.724	1.396	1.245
AgACI + squared	0.387	0.102	0.413	0.130
DtACI + abs	0.774	0.735	1.434	0.822
DtACI + cdf-tail	0.860	0.876	1.262	1.225
DtACI + cqr	0.802	0.734	1.388	1.300
DtACI + diff	0.799	0.745	2.142	1.833
DtACI + distributional	0.828	0.843	1.319	1.510
DtACI + iqr-scaled	0.785	0.735	1.95×10^7	1.76×10^8
DtACI + log	0.819	0.792	2.215	2.435
DtACI + mad-scaled	0.796	0.724	1.432	1.244
DtACI + scaled-cqr	0.822	0.730	1.462	1.269
DtACI + signed	0.796	0.724	1.435	1.245
DtACI + squared	0.389	0.102	0.418	0.132
Native Quantiles	0.782	0.795	1.000	1.000
PID + abs	0.720	0.627	1.183	0.610
PID + cdf-tail	0.835	0.861	1.150	1.136
PID + cqr	0.745	0.598	1.230	0.689
PID + diff	0.700	0.516	1.92×10^{285}	1.55×10^{286}
PID + distributional	0.758	0.749	1.082	1.208
PID + iqr-scaled	0.732	0.633	8.36×10^6	7.12×10^7
PID + log	0.685	0.598	1.836	1.857
PID + mad-scaled	0.692	0.512	1.156	0.805
PID + scaled-cqr	0.738	0.534	1.221	0.805
PID + signed	0.692	0.512	1.156	0.805
PID + squared	0.373	0.096	0.397	0.125
SplitCP + abs	0.712	0.558	1.321	1.002
SplitCP + cdf-tail	0.839	0.861	1.167	1.151
SplitCP + cqr	0.757	0.549	1.384	0.910
SplitCP + diff	0.685	0.456	1.770	1.176
SplitCP + distributional	0.754	0.727	1.082	1.224
SplitCP + iqr-scaled	0.723	0.555	1.06×10^7	9.87×10^7
SplitCP + log	0.650	0.527	2.043	2.023

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Table A7 – Continued from previous page

Calibration Method	Coverage $\uparrow$		Scaled Width $\downarrow$	
	Task	Series	Task	Series
SplitCP + mad-scaled	0.668	0.440	1.179	0.766
SplitCP + scaled-cqr	0.738	0.490	1.250	0.755
SplitCP + signed	0.668	0.440	1.179	0.766
SplitCP + squared	0.375	0.097	0.406	0.129
TrailingWindow + abs	0.722	0.698	1.032	0.709
TrailingWindow + cdf-tail	0.827	0.850	1.131	1.105
TrailingWindow + cqr	0.745	0.690	1.107	1.024
TrailingWindow + diff	0.724	0.701	1.462	1.465
TrailingWindow + distributional	0.764	0.770	0.980	0.997
TrailingWindow + iqr-scaled	0.737	0.694	1.63×10^7	1.49×10^8
TrailingWindow + log	0.737	0.712	1.103	1.124
TrailingWindow + mad-scaled	0.715	0.682	0.987	0.986
TrailingWindow + scaled-cqr	0.748	0.683	1.089	0.990
TrailingWindow + signed	0.715	0.682	0.985	0.967
TrailingWindow + squared	0.358	0.095	0.339	0.107
WeightedCP + abs	0.803	0.826	1.356	0.998
WeightedCP + cdf-tail	0.842	0.876	1.190	1.216
WeightedCP + cqr	0.825	0.844	1.421	1.695
WeightedCP + diff	0.825	0.853	1.998	2.398
WeightedCP + distributional	0.810	0.868	1.182	1.372
WeightedCP + iqr-scaled	0.814	0.833	1.39×10^7	1.58×10^8
WeightedCP + log	0.844	0.876	1.981	2.447
WeightedCP + mad-scaled	0.821	0.852	1.348	1.649
WeightedCP + scaled-cqr	0.839	0.866	1.413	1.709
WeightedCP + signed	0.821	0.852	1.348	1.649
WeightedCP + squared	0.438	0.166	0.440	0.198